

Clinical Differential Diagnosis and Variants of Actinic Keratosis

Lichen planus-like keratosis and **Bowen disease** must be considered in the differential diagnosis of actinic keratosis (AK), particularly in solitary lesions. Clinical variants of AK include **pigmented**, **Bowenoid**, and **lichenoid** forms. While Bowenoid and lichenoid AK are typically diagnosed via histopathologic examination, **pigmented AK** commonly arises in sun-damaged skin, necessitating differentiation from **lentigo maligna**.

Important clinical clues for diagnosing AK include: - Number and location of lesions

- Similar morphology across a region (comparative approach)
- Rough surface on palpation

Clinical signs suggestive of progression to **invasive squamous cell carcinoma (iSCC)** include: - Induration

- Bleeding
- Discomfort or pain
- Increase in thickness and diameter

Actinic keratoses rarely appear as solitary lesions; instead, they often affect an entire anatomical region, reflecting field cancerization.

Olsen Classification of AK

Individual AK lesions are clinically graded by thickness using the **Olsen classification**: - **Grade 1**: Slightly palpable, more easily felt than seen

- **Grade 2**: Moderately thick, easily seen and felt
- **Grade 3**: Very thick and hyperkeratotic

This grading system has been combined with lesion counting in defined areas for clinical trial assessments. However, lesion counting lacks reproducibility, even among experts. Although the Olsen classification does not reliably

correlate with histological severity, it has been shown to strongly predict the risk of developing cutaneous squamous cell carcinoma (cSCC).

Other clinical severity indexes have been proposed that assess the overall affected area rather than individual lesions.

Disease Burden-Based Classification

A recent clinical practice approach classifies AK based on overall disease burden into four categories:

1. **Single AK:** Fewer than 5 lesions in a defined field

2. **Multiple AKs:** 6 or more lesions in a defined field

3. **Field cancerization:** 6+ lesions with sun-damaged skin and hyperkeratosis

4. **AKs associated with immunosuppression**

Dermatoscopy (Table 3)

Dermatoscopy enhances the clinical diagnosis of AK, with reported sensitivity and specificity of **98.7%** and **95.0%**, respectively. Dermoscopic findings vary by clinical grade:

- **Grade 1 AK:** Red network pattern
- **Grade 2 AK:** "Strawberry pattern" — background erythema with white-colored follicular openings, sometimes filled with keratin plugs
- **Grade 3 AK:** Structureless white to yellow areas

Dermatoscopy aids in differentiating: - **Solitary AK grade III** (structureless yellow-white areas, focal strawberry pattern) from **Bowen's disease** (coiled vessels)

- **Pigmented AK** from **lentigo maligna**, based on: - Enlarged white follicular openings

- Double lines between hair follicles

- Background erythema

- Scaly surface

In non-facial locations, follicular findings are less prominent; dermatoscopy typically shows erythema and superficial scales.

Dermatoscopy also helps detect early signs of **iSCC**, such as: - Coiled or glomerular vessels

- Polymorphous vessels
- White circles

These features are rarely seen in typical AK.

Confocal Microscopy and Optical Coherence Tomography (OCT) (Table 4)

Reflectance confocal microscopy (RCM) and optical coherence tomography (OCT), when used alongside clinical examination and dermatoscopy, can assist in the differential diagnosis of: - **AK vs. SCC**

- **Pigmented AK vs. lentigo maligna**